

LighterBody: RNN based Anticipated Virtual Body Makes You Feel Lighter

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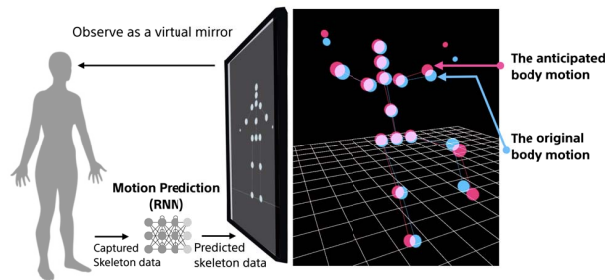


Figure 1: LighterBody, a system for real-time temporal manipulation of a virtual body to induce changes in the feeling of the bodyweight. By showing a temporally anticipated virtual body representation in the screen, users feel change in their own bodyweight sensation.

ABSTRACT

The virtual body representation had shown the potential of intervention into the sense of body. To investigate how the temporal shift of body representation affects the user's kinetic sensation, we developed a system to anticipate body movement with RNN. We then conducted a user study to assess the effect with the anticipated body movement and the system baseline. Results revealed that the transition from the baseline to the anticipated body induced a lighter bodyweight feeling, and the opposite transition induced a heavier feeling. Our work enlightens the potential of interactive manipulation of the full-body kinetic sensation using virtual body representation.

1 INTRODUCTION

We understand our movements during physical exercise by looking at our reflection in a mirror in addition to the internal proprioceptive feedback [1]. Visual pseudo haptics research has shown that it is possible to generate a virtual snag or weight sensation by controlling the delay distance to the visual representation and the control display ratio [2, 4]. Then, if the delayed intervention of the body representation makes the body feel heavier, does the presumed prior display of the body representation make the body feel lighter? Such changes in the body sensation have been reported with simple motion prediction [5]. Makino et al. also demonstrated that displaying a silhouette with the predicted center of the body induces the change of bodily feeling [7]. Here we envision that the sophisticated method of full body pose prediction allows us detailed investigation with various human movements. Therefore, in this study, we constructed a prediction network using RNNs to anticipate body movement, and examined whether the visual exposure of the anticipated body pose can alter the kinetic perception during the body movement.

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2 LIGHTERBODY: REAL-TIME BODY POSE ANTICIPATION SYSTEM

LighterBody system consists of three components; Kinect V2 sensor to acquire real-time users skeleton data (25 joints with position information, 18 bones with rotational information, captures in 30Fps), a low-latency LCD display (1920 x 1080, used in 120Hz, input latency 5ms) and a computer (Ryzen 3960X CPU and GTX 2070 Super GPU) for processing including pose estimation and rendering the graphics. We measured the system latency from capturing the bone to displaying the graphics by comparing the actual motion and the graphics with high frame rate camera, and found the system latency is min. 66ms. In this experiment, we chose the configuration of the virtual mirror displayed on display without covering participants' eyes and faces. We can assume this setup would be one of the possible alternatives to the situation where the virtual mirror is placed on Virtual Reality.

2.1 RNN based body pose prediction

Here, we used QuaterNet [8], a model that is trained on a quaternion basis and allows for a high-precision estimate. The QuaterNet allows us to configure the target length of the prediction. Following the research, we used human3.6M data set [3] for training data. We used the 19 joints rotational information from Kinect V2 for real-time prediction. Since some of the rotational information were observed as quite unstable, we recalculated them based on neighbor joints' position data with the prior constraints of human bones. Because of the same reason, rotational information of foot and hand were excluded.

To make a consistent structure between the skeleton data obtained from Kinect V2 and the human 3.6M skeleton structure, we performed a transformation of the skeleton data. We reduced the number of joints of human3.6M from 32 to 19 joints. In addition, the Human 3.6M data-set was recorded at 50 Hz, so we performed down-sampling the data stream of Human3.6M with a slerp interpolation to match the Kinect v2 frame-rate (30 Hz). When training the model, the input size for RNN was also changed from 32×4 to 19×4 because of the change of the skeleton structure. We configured the maximum target length as 10 frames, which is 333ms in 30 Fps, from the accumulation of the latest 50 frames data. Other parameters and architectures are the same as QuaterNet. Throughout the training, the batch size was set to 60, and the number of epochs was 1000. For the training data, we selected 15 actions from 5 subjects in human 3.6M data set.

With consideration of the trade-off between the length of anticipation and the prediction error, we optimized parameters of RNN to achieve approx. 100 ms anticipation with approx. RMS 40(mm) prediction error.

3 USER STUDY

In this study, our aim was to investigate whether the temporal anticipation in the body representation affects the kinetic sense of users. To capture the sense of change in participants' kinetic sense, we focus on subjective reports of the sense of bodyweight.

In our experiments, we used simple points and lines as the body representation in the virtual mirror in order to eliminate the influence of the realism of the virtual body, texture, and avatar representation

format and to force participants to focus only on the motion information [6].

This study investigates the sensory change in body weight when the body expression is switched from the PRE state to the POST state. We present PRE state for 10 seconds, and then POST state for 4 seconds. The reason presentation time of PRE state is longer than that of POST state is to take time for participants to get used to the PRE state. We asked participants to perform periodic movement (the moderate speed cyclic motion) during the trial. Participants swing their arm vertically and slightly move the trunk vertically, similar to a squat movement. The cycle of the instructed movement is approx. 0.5 Hz. We set up this exercise pattern to provide sufficient exercise intensity and yet avoid extreme fatigue. To avoid participants seeing their own body, participants wear the head-mounted tool without a lens to limit the field of view so that participants can only see the display.

We designed a user study with four combinations of PRE and POST conditions: **1) Baseline (of the system) to Anticipation**, **2) Anticipation to Baseline**, **3) Baseline to Baseline (no change)**, and **4) Anticipation to Anticipation (no change)**

In each trial, participants report the change of bodyweight sensations they felt during the exercise on 5 points Likert scale (1 : Lighter - 5: Heavier).

We recruited 11 participants (five self-identified as female, six as male) from a local institute. Their ages ranged from 23 to 40 years (mean = 27.0, SD = 4.0). The study design was approved by the local ethics committee and all participants provided written informed consent prior to their participation. All had a normal or corrected-to-normal vision. Five of the participants had a regular exercise in daily life, and others had less exercise.

3.1 Result

In 4 possible condition, we acquired 10 samples of data points and then calculated the median score in each condition for each participants. To investigate the significant difference in switching the skeleton mode, we subtracts the median value in the (baseline-baseline) condition from other three condition for each participants. Figure 2 show the relative score from 3) baseline-baseline condition.

The Friedman tests was conducted to compare the effect of the condition on the relative score among three condition. There was a significant difference among the distributions of the three condition (based on Friedman's test, $\chi^2(2) = 13.8, p < .005$). Subsequent post-hoc, Wilcoxon signed-rank tests revealed that there were significant difference ($p < 0.05$) between the conditions between 1) Baseline - Anticipated and 2) Anticipated - Baseline. From this analysis, we found that switching the virtual body representation (baseline - anticipated) induces the change of sensation on bodyweight.

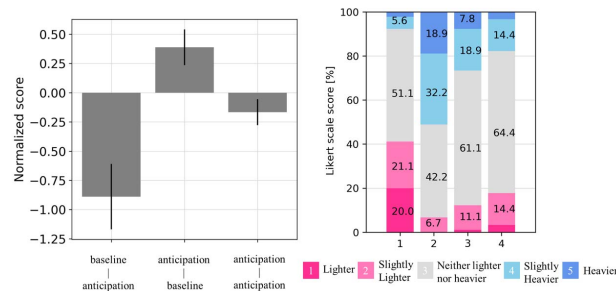


Figure 2: Left: The means of the relative scores from 3) baseline-baseline condition. Bar indicates SE. Right : Each participant's response. Bar indicates the percentage of 5 points Likert scale score in four conditions.

3.2 Discussion

Looking at the results for each participants, we found that the trend of subjective response was clearly different between the group that reported the sensory change and the group that did not.

We examined and cross-referenced these trends with post-experimental interviews, as well as participants' ages and exercise experiences. Interestingly, in our experiment, participants who responded to clear sensory changes were older and less routinely exercised than those who did not. It is also interesting to note that those participants who had significant reports of fatigue at task completion (P1,2, 5, and 6) responded with clear sensory changes. This suggests that the intensity of the sensory changes may be influenced by the subjects' exercise experience, physical fitness, or the degree of fatigue.

In addition, whether participants could project themselves into the displayed skeleton visualization is also assumed to be an essential factor. As P5 mentioned, 'Sometimes, I saw point shaking in the displayed skeleton' in the post-interview, those visual noises would prevent projecting themselves into the skeleton visualization. Therefore, the stability of the movement of the virtual graphics can be affect the subjective response.

4 CONCLUSION

We presented LighterBody, a real-time system to manipulate the virtual body to induce changes in the feeling of the bodyweight. Our RNN-based prediction system can anticipate approx. 100 ms from the system baseline. Our user study revealed that switching the virtual body representation from the baseline to the anticipated body induced a lighter bodyweight feeling, and the opposite transition induced a heavier feeling. Our work enlightens the potential of interactive manipulation of the full-body kinetic sensation using virtual body representation.

ACKNOWLEDGMENTS

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